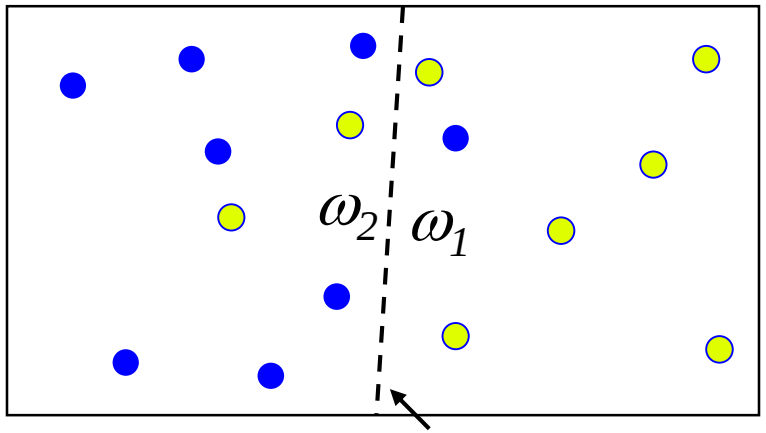
Classifier Evaluation

Classifier Evaluation

The most simple way to evaluate a classifier is to see the number of correct and incorrect classifications.

For a M-class problem with N samples to be classified, the **confusion matrix** is a MxM matrix, whose (i,j) element is the number of vectors that are actually in class ω_i and classified to class ω_j.

Example (2-class): ω_1 ω_2



Confusion Matrix: $\begin{pmatrix} 6 & 2 \\ 1 & 7 \end{pmatrix}$

Correct classification rate
= trace(Confusion Matrix)/N

Confusion Matrix Examples

A typical confusion matrix for the Iris dataset:

50	0	0
0	46	4
0	0	50

Clothing color classification:



Real \ Pred	Red	Orange	Yellow	Green	Blue	Pink	Purple	Brown	Gray	Black	White
Red	167	17	1	0	4	23	8	4	3	9	2
Orange	4	37	13	0	2	0	0	2	0	1	0
Yellow	3	1	87	5	0	3	0	5	3	1	3
Green	0	0	9	100	7	2	0	3	8	8	3
Blue	0	0	0	13	450	10	6	0	42	114	21
Pink	16	2	2	0	2	124	6	3	5	2	9
Purple	9	0	1	1	23	21	70	1	7	15	2
Brown	3	2	8	12	0	7	0	66	14	22	7
Gray	4	0	1	23	21	15	1	14	289	38	38
Black	10	1	0	15	44	15	15	5	49	903	9
White	1	0	2	7	29	26	2	4	52	9	322

Two-Class Confusion Matrix

Many two-class problems can be considered as "detection" problems where the classifier is expected to answer a "Yes/No" question for each sample, such as in a medical screening test.

Let class#1 be "No", class#2 be "Yes", confusion matrix be $\begin{pmatrix} TN & FP \\ FN & TP \end{pmatrix}$

Common metrics (in pairs) derived from the confusion matrix:

PD (probability of correct detection) = $TP / (TP + FN)$
FA (probability of false positive/alarm) = $FP / (TN + FP)$
Recall = **PD**
Precision = $TP / (TP + FP)$
Sensitivity = **PD**
Specificity = $TN / (TN + FP) = 1 -$ **FA**
PPV (positive predictive value) = **Precision**
NPV (negative predictive value) = $TN / (TN + FN)$

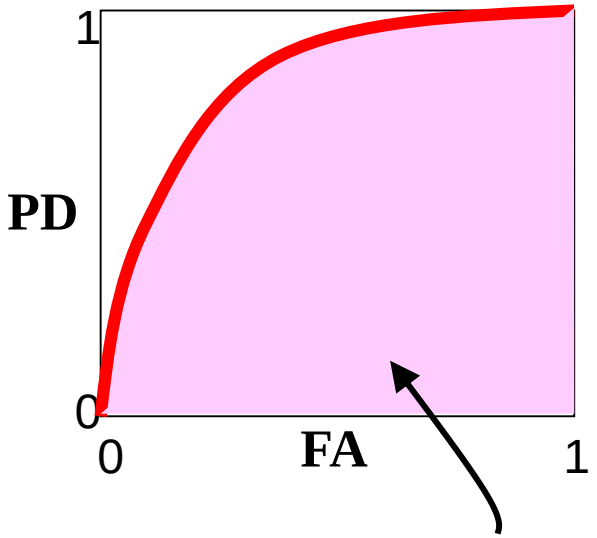
Two-Class Confusion Matrix

- A threshold can be used to adjust how sensitive the classifier is to identify positive cases.
- This adjustment, while increasing one metric in a pair (e.g., **Recall**), is likely to decrease the other (e.g., **Precision**).
- **F1 measure** is one combined metric to allow for easier comparison between such paired classification results. It can also be used to select a “proper” threshold.

$$F1 = \frac{2 * precision * recall}{precision + recall}$$

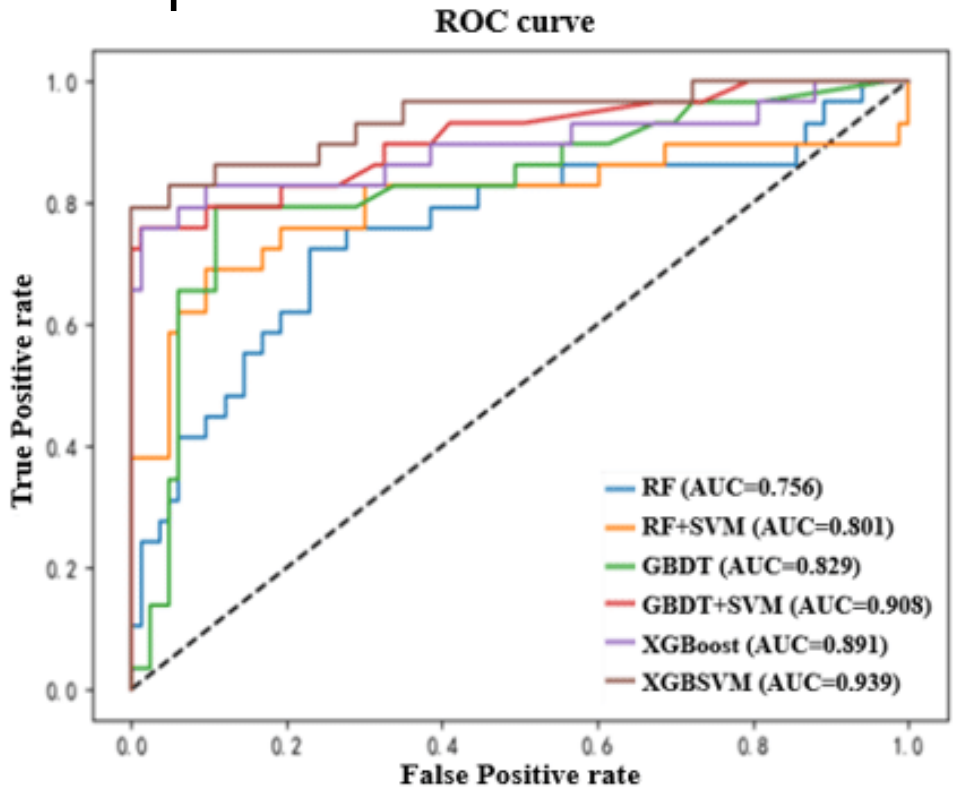
ROC Curves

Receiver Operating Characteristics (ROC) Curve is the plot of **PD** vs. **FA** at different threshold values. It allows the evaluation of different classification methods, settings, and/or features without choosing the thresholds.



AUC or **AUROC** (area under the ROC curve): The larger, the better.

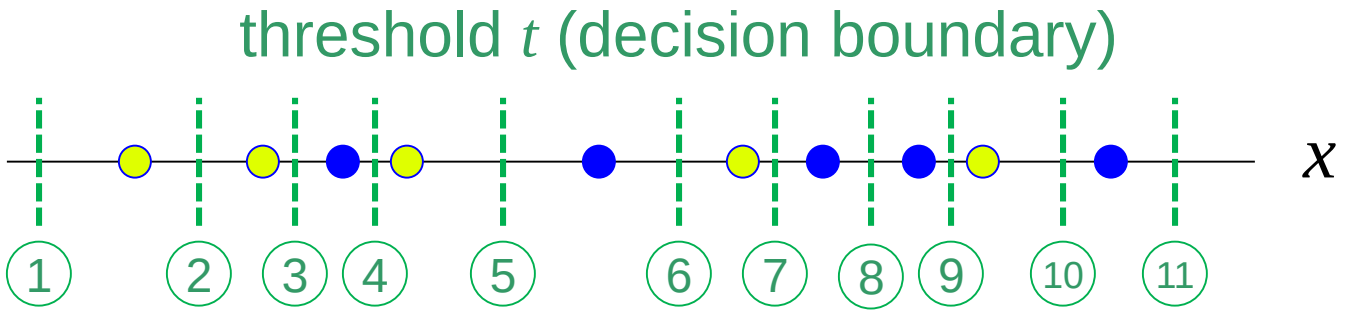
Example:



ROC Curves

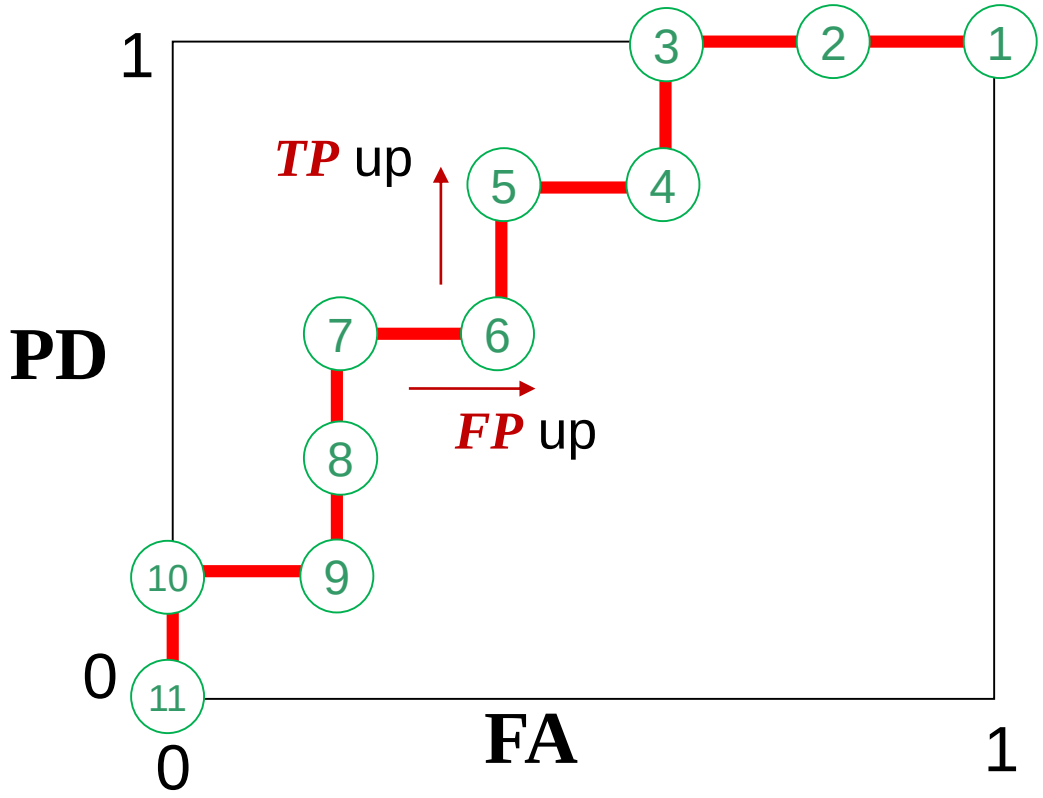
Example (1-D):

negative positive



PD = $TP / (TP + FN)$

FA = $FP / (FP + TN)$

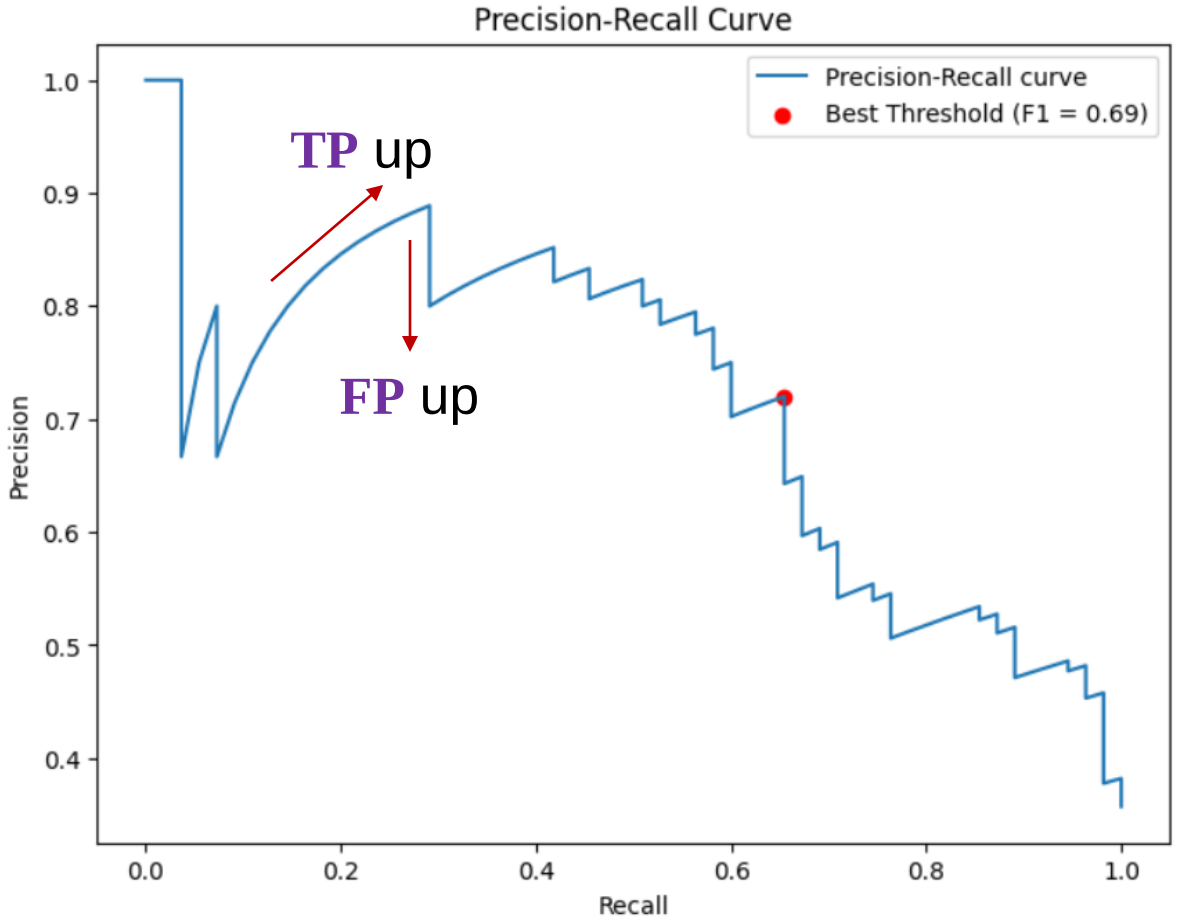


Precision-Recall (PR) Curves

Recall = $TP / (TP + FN)$

Precision = $TP / (TP + FP)$

Neither depends on **TN**, so PR curves are suitable for cases when **TN** is undefined, such as detection problems. The area under the curve (0~1) is usually called **average precision** (**AP**).



https://www.blog.trainindata.com/precision-recall-curves/

Dataset Division

- It is important that the labeled data for evaluation is separated from the labeled data used for training / model parameter estimation.
- The concept of validation: To estimate the performance of the model on previously unseen data.
- It's often that the majority of labeled data is used for training to get reliable models.
- The distributions of the training and validation/testing subsets are as similar as possible.
- The samples in the training and validation/testing subsets should be unrelated / uncorrelated when possible.
- With limited amount of labeled samples, cross-validation is commonly used to obtain reliable performance estimation.

Cross-Validation

- The N labeled samples are divided into K subsets of approximately equal sizes ($K>1$). They should have similar distributions.
- The training process (model parameter estimation) is run K times (K trials).
- In the Kth trial, the Kth subset is used for validation, and the other subsets are used for training.
- The overall performance is the combination of the performance on all the validation subsets.
- Extreme case: The leave-one-out method ($K=N$).

Illustration of 3-fold cross-validation:

parameter estimation | validation

Trial 1:	#1	#2	#3
Trial 2:	#1	#2	#3
Trial 3:	#1	#2	#3

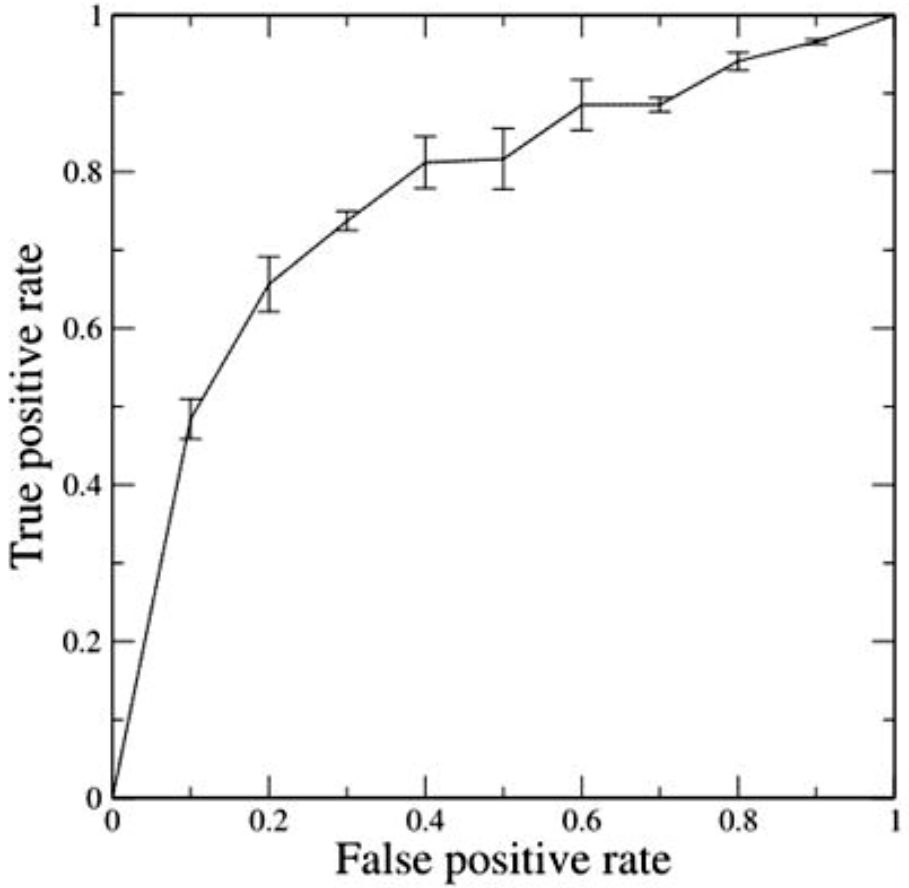
Classifier Evaluation w/ Cross-Validation

- Confusion matrix: Add the validation-subset confusion matrices from all the trials. Each sample is included exactly once. Metrics based on the confusion matrix, such as **PD** and **FA**, can then be computed.
- ROC curve (two-class problems):
  - Method 1: Just collect the outputs of the validation subsets from all the trials and draw a single ROC curve.
  - Method 2: Compute a ROC curve for each trial. We can average the curves at any given **FA** rate. The standard deviation gives us an estimation of the uncertainty of **PD** at any **FA** rate.
  - Method 3: Use a common set of thresholds to compute separate **PD** and **FA** values for all the validation subsets. This produces a single ROC curve with uncertainties for both **PD** and **FA**.

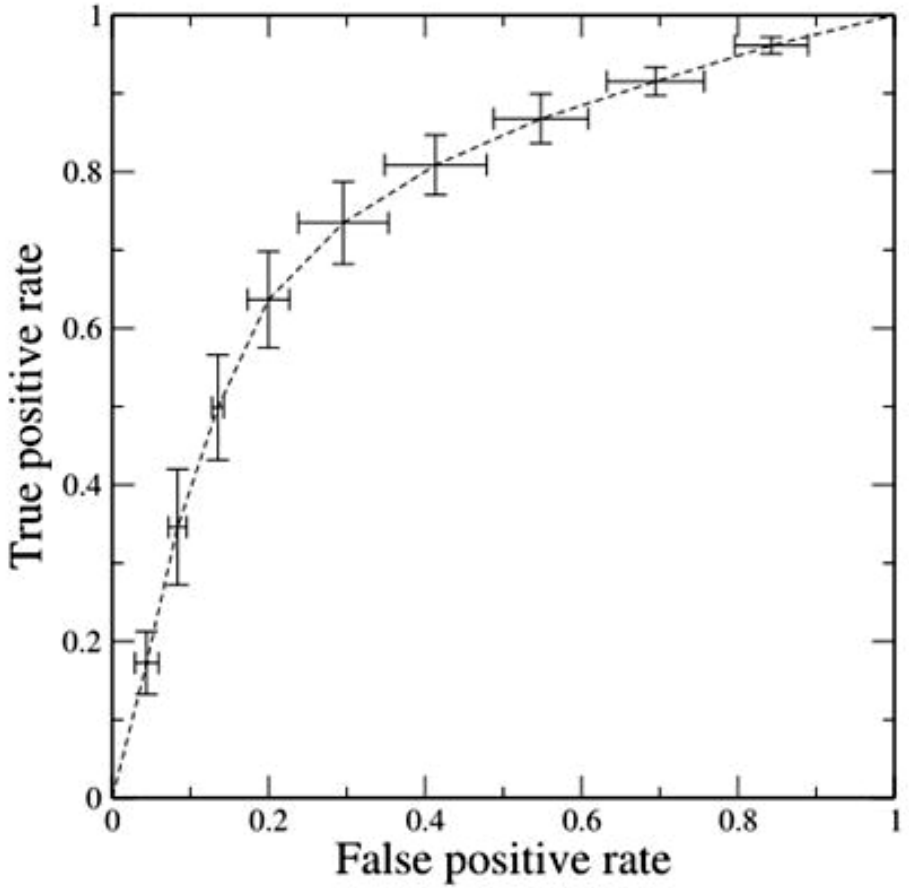
Classifier Evaluation w/ Cross-Validation

Examples of generating ROC curves with cross-validation:

(method 2)



(method 3)



https://slideplayer.com/slide/7028153/

Classification for Imbalanced Datasets

- Imbalanced datasets: Datasets where different classes have different proportions of samples.
- Without extra care, classifiers trained with imbalanced datasets tend to generate outputs that favor the majority class(es).
- Techniques for dealing with imbalanced datasets:
  - Oversampling the minority class and/or under-sampling the majority class
  - Class weighting
  - Generating extra samples for the minority class through synthesis, data augmentation, etc. (related to oversampling of the minority class)
  - Some classifier architectures are less sensitive to imbalanced data.

Classification for Imbalanced Datasets

Examples of imbalanced datasets:

- ➔ ➔
- Anomaly detection:
  - Medical diagnosis
  - Fraud/spam detection
  - Accident/disaster detection
  - Defect detection
  - …
- How/where/when you do the dataset collection…

Clothing color classification:

Real \ Pred	Red	Orange	Yellow	Green	Blue	Pink	Purple	Brown	Gray	Black	White
Red	167	17	1	0	4	23	8	4	3	9	2
Orange	4	37	13	0	2	0	0	2	0	1	0
Yellow	3	1	87	5	0	3	0	5	3	1	3
Green	0	0	9	100	7	2	0	3	8	8	3
Blue	0	0	0	13	450	10	6	0	42	114	21
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Brown	3	2	8	12	0	7	0	66	14	22	7
Gray	4	0	1	23	21	15	1	14	289	38	38
Black	10	1	0	15	44	15	15	5	49	903	9
White	1	0	2	7	29	26	2	4	52	9	322

Classifier Evaluation for Imbalanced Dataset

Some metrics may be misleading if the classes are imbalanced, even for two-class problems. Example: Consider the following two-class confusion matrix:

	F	T
F	1	10
T	10	100

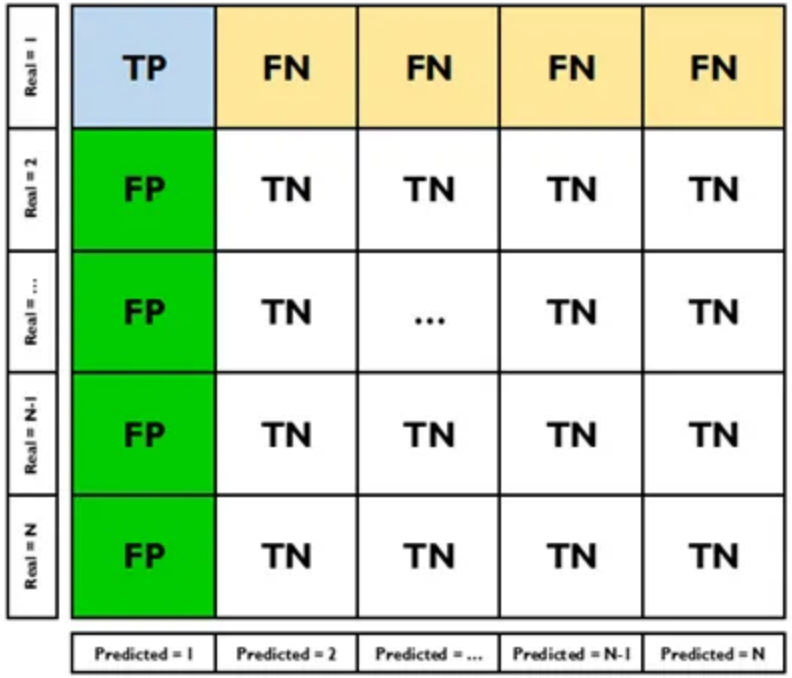
Based on the predefined positive and negative classes, the recall, precision, and F1 score are all 0.9, which seem very good.

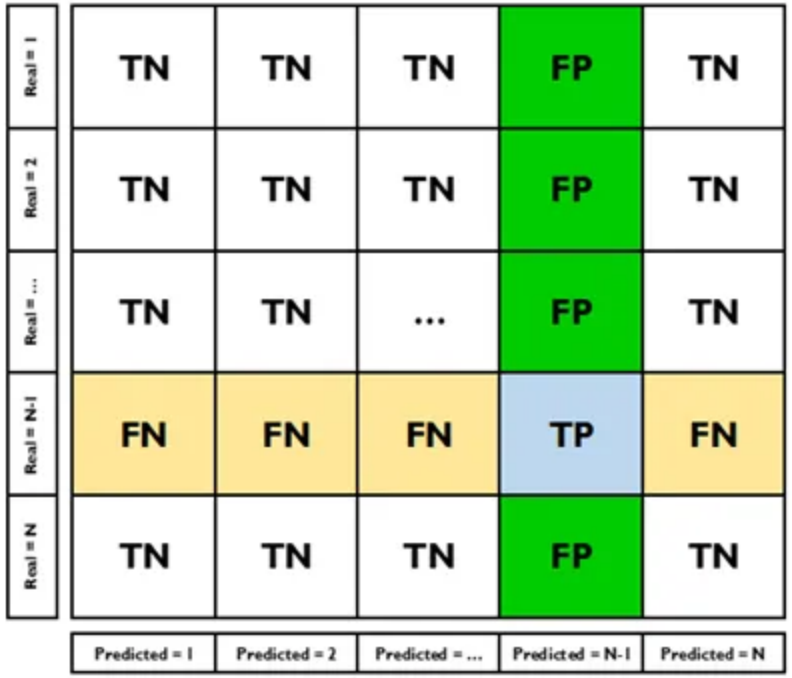
However, this classifier is actually doing random guess based on the relative amount of training data in each class.

We need to consider using all the elements in the confusion matrix for evaluation, such as using the specificity/sensitivity pair, or macro-F1 (to be discussed).

Evaluation for Multi-Class Classification

- Q: Other than accuracy, how are the other metrics extended to multi-class problems?
- The metrics for binary classifiers can be computed in a per-class **one-vs-all** basis:





https://medium.com/mcd-unison/multiclass-confusion-matrix-clarity-without-confusion-88af1494c1d1

Evaluation for Multi-Class Classification

- The evaluation metrics of all the classes can be combined to obtain the overall evaluation metrics.
- Example: mAP (mean AP) is the average AP of the different classes in multi-class object detection problems.
- **Micro-average**:
  - The per-class binary confusion matrix components (TP, FP, etc.) are summed over the classes before computing metrics.
  - Each item in the data contributes equally regardless of its class; micro-precision and micro-recall becomes the same as overall accuracy.
- **Macro-average**:
  - Metrics (PD, FA, etc.) are averages of their per-class (one-vs-all) values.
  - Macro-F1 is computed from macro-averaged precision and recall.
  - Each class contributes equally.

Evaluation for Multi-Class Classification

An example of multi-class ROC curves from scikit-learn:

